

1.Brainstorming &Ideation Phase

Project Title:

Script Controlled ACL-Restricted Record Access Based on Field Value

Objectives

The major objectives of the project are:

- To understand the concept of Access Control Lists.
- To implement a script-based ACL.
- To restrict access based on a field value.
- To provide record-level security.
- To prevent unauthorized users from viewing restricted records.
- To make access control dynamic and condition-based.
- To reduce unnecessary exposure of application data.

Problem Identification

In a normal access-control system, permissions may be provided based mainly on roles. However, role-based access alone may not be sufficient when access needs to depend on the information stored in an individual record.

For example, consider a table containing records with a field called Department. A user may have permission to access the table, but the system should allow the user to view only records belonging to their department.

Without field-based access control:

- Users may view records that are not related to their work.
- Sensitive information may be exposed unnecessarily.

- It becomes difficult to implement record-level security.
- Manual checking of records increases administrative effort. Therefore, a dynamic access-control mechanism is required.

Idea Generation

The main idea generated during brainstorming is to create a script-controlled ACL that checks a selected field value before granting access to a record.

The basic concept is: User → Request Record → ACL Executes → Field Value Checked → Access Allowed/Denied

For example, if the record contains a field called Department, the ACL script can compare the record's department with the user's department. If the values match, access can be granted. If they do not match, access can be restricted.

This approach provides more flexible and dynamic access control than relying only on static roles.

Proposed Solution

The proposed solution is to configure an ACL for the required table and use a script to evaluate the value of a specific field.

The ACL will perform the following basic operations:

1. Identify the user requesting access.
2. Access the requested record.
3. Read the required field value from the record.
4. Compare the field value with the required condition.
5. Grant access when the condition is satisfied.
6. Deny access when the condition is not satisfied.

This provides an additional security layer at the record level.

Scope of the Project

The project focuses on implementing field-value-based access control for records. The solution can be adapted to different tables and different fields depending on the application requirements.

The ACL can be designed to check fields such as:

- Department
- Location
- Assignment group
- Category
- Status
- User-related fields

The project mainly concentrates on controlling access to records rather than modifying the complete application security architecture.

Expected Outcome

The expected outcome is a working ACL configuration in which record access is controlled by a script based on a selected field value.

When a user requests a record, the ACL should evaluate the condition automatically. If the condition is satisfied, the user can access the record. Otherwise, access is restricted.

Thus, the project demonstrates how scripting can be combined with ACLs to provide flexible and record-level security.

1.8 Conclusion

During the brainstorming and ideation phase, the need for more flexible record-level security was identified. The idea of using a script-controlled ACL was selected because it can dynamically evaluate field values before allowing access.

The proposed project provides a simple approach to demonstrate how access to records can be controlled according to specific conditions. It also provides a foundation for implementing more advanced access-control rules in future applications.